Kaiyu Yang

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I lead a research team at Apodex working on the mathematical capabilities of language models and on verification. AI systems now generate code, proofs, and arguments faster than anyone can check them, and formal systems remove that bottleneck: a solution arrives with a specification and a machine-checkable proof. I have worked on machine learning for formal mathematics since 2018, from early benchmarks (CoqGym) to open-source platforms (LeanDojo, Lean Copilot) and open provers (Goedel-Prover).

Experience

Apodex, Redwood City, CA — Mar 2026 – Present
Lead Scientist

- Lead a research team on the mathematical capabilities of LLMs, working across the training stack rather than at a single stage — data, mid-training, post-training, and agents — since no one stage creates mathematical ability and each bounds what the next can add.
- Second workstream on verification: formal guarantees for the infrastructure that training runs on, and verifiers that tell an agent whether it is still on track, which matters more as tasks lengthen and errors compound.
- Direct the research agenda and external academic collaborations on theorem proving and verified code generation.

Meta Fundamental AI Research (FAIR), New York, NY — Jun 2024 – Dec 2025
Research Scientist

- Co-authored *Formal Mathematical Reasoning: A New Frontier in AI* (ICML 2025 position paper; also in *Communications of the ACM*), which lays out the research agenda connecting LLMs with formal methods.
- Advised the Goedel-Prover series (COLM 2025, ICLR 2026) — open-source theorem provers that set the state of the art among open models — and ProofOptimizer (ICLR 2026), a model that simplifies machine-generated proofs without human demonstrations.
- Collaborations on verified code generation (Verina, ICLR 2026) and neuro-symbolic olympiad problem solving (ICLR 2025, CAV 2025).

California Institute of Technology, Pasadena, CA — Sep 2022 – May 2024
Postdoctoral Fellow

- Built LeanDojo (NeurIPS 2023), an open-source platform for LLM-based theorem proving in Lean, and Lean Copilot, which runs LLM inference natively inside the Lean proof assistant; both are in regular use in the field.

Education

Princeton University, Princeton, NJ — 2022
PhD in Computer Science

University of Michigan, Ann Arbor, MI — 2018
MS in Computer Science and Engineering

Tsinghua University, Beijing, China — 2016
B.Eng. in Computer Science and B.S. in Mathematics

Selected Publications

Full list on Google Scholar: 30+ papers, over 10,000 citations (NeurIPS, ICML, ICLR, COLM, CACM, EMNLP, CVPR, ICCV, ECCV, CAV).

- **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**
  K. Yang, A. Swope, A. Gu, R. Chalamala, P. Song, S. Yu, S. Godil, R. Prenger, A. Anandkumar. NeurIPS 2023.
- **Formal Mathematical Reasoning: A New Frontier in AI**
  K. Yang, G. Poesia, J. He, W. Li, K. Lauter, S. Chaudhuri, D. Song. ICML 2025; Communications of the ACM.
- **Goedel-Prover: A Frontier Model for Open-Source Automated Theorem Proving**
  Y. Lin*, S. Tang*, B. Lyu, J. Wu, H. Lin, K. Yang, J. Li, M. Xia, D. Chen, S. Arora, C. Jin. COLM 2025.

- **Goedel-Prover-V2: Scaling Formal Theorem Proving with Scaffolded Data Synthesis and Self-Correction**
  Y. Lin*, S. Tang*, B. Lyu*, Z. Yang*, et al. ICLR 2026.
- **Goedel-Code-Prover: Hierarchical Proof Search for Open Code Verification**
  Z. Li*, Z. Yang*, D. He, H. Zhao, A. Zhao, S. Tang, K. Yang, A. Gupta, Z. Su, C. Jin. COLM 2026.
- **ProofOptimizer: Training Language Models to Simplify Proofs without Human Demonstrations**
  A. Gu, B. Piotrowski, F. Gloeckle, K. Yang, A. Markosyan. ICLR 2026.
- **Verina: Benchmarking Verifiable Code Generation**
  Z. Ye, Z. Yan, T. Kasriel, J. He, K. Yang, D. Song. ICLR 2026.
- **Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean**
  P. Song, K. Yang, A. Anandkumar. NeuS 2025.
- **A Survey on Deep Learning for Theorem Proving**
  Z. Li, J. Sun, L. Murphy, Q. Su, Z. Li, X. Zhang, K. Yang, X. Si. COLM 2024.
- **Proving Olympiad Inequalities by Synergizing LLMs and Symbolic Reasoning**
  Z. Li*, Z. Li*, W. Tang, X. Zhang, Y. Yao, X. Si, F. Yang, K. Yang†, X. Ma†. ICLR 2025.
- **Generating Natural Language Proofs with Verifier-Guided Search**
  K. Yang, J. Deng, D. Chen. EMNLP 2022.
- **Learning to Prove Theorems via Interacting with Proof Assistants**
  K. Yang, J. Deng. ICML 2019.
- **Stacked Hourglass Networks for Human Pose Estimation**
  A. Newell, K. Yang, J. Deng. ECCV 2016. Earlier computer-vision work; carries most of the citation count above.

Open Source

Core developer of open-source tooling for machine learning with the Lean and Coq proof assistants:

- LeanDojo — interacting with Lean from machine-learning pipelines; 800+ GitHub stars.
- ReProver — retrieval-augmented theorem prover; 300+ stars.
- Lean Copilot — LLM inference inside Lean for proof automation; 1,300+ stars.
- CoqGym — learning environment for the Coq proof assistant; 400+ stars.
- LeanDojo models on Hugging Face — retrieval and tactic-generation models; about 7,000 downloads a month.

Service and Community

- Co-organizer, Workshop on Mathematical Reasoning and AI (MATH-AI) at NeurIPS 2023, 2025, and 2026; co-organizer of the NeurIPS 2023 tutorial on machine learning for theorem proving.
- Area Chair: ICML 2025 & 2026, NeurIPS 2026, ECCV 2024. Reviewer for NSF panels and ERC Advanced Grants.
- Guest co-instructor, *Advanced Large Language Model Agents* (UC Berkeley & MOOC, 2025); guest lecturer at Caltech and CUHK.
- Invited talks at OpenAI, Google, Meta FAIR, the Simons Institute, UC Berkeley, Stanford, CMU, and other universities.
- Mentored students and junior researchers who went on to PhD programs at Toronto, Cornell, CUHK, and UIUC, and to industry labs.
- Work on AI for mathematics covered in *Scientific American* (2024).

* equal contribution   † equal advising